From Vision-Language-Action to Motion:
Mapless Navigation: LiDAR-Guided Arc Avoidance

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Motivation

Indoor robots usually navigate on a prebuilt map, which takes a survey to build and goes out of date as soon as objects move. A **vision-language-action (VLA) policy** needs no map. From one camera image and a prompt instruction it is able to generate waypoints directly. But it is trained on clean demonstrations and has no understanding of free space, so some things can go wrong:

- **Turning.** The policy undershoots, so the path cuts into the inside corner of the junction.
- **Obstacles.** A chair or bin in the corridor is not part of what the policy learned, so the path runs through it.
- **Other people.** Someone walking toward the robot is a moving constraint the policy cannot react to.

We check the policy-derived arc against single-scan LiDAR obstacle points. Clearance checks adjust speed and steering, and no accumulated map is used.

Objectives

1. **Map-free policy:** turn one camera frame and a standing instruction into a waypoint trajectory, on the robot itself.
2. **Single-scan geometry:** estimate centring and openings without an accumulated map; filter scene labels across scans.
3. **Stable tracking:** use pure pursuit with an additive LiDAR PD centring correction in corridors.
4. **Obstacle clearance:** choose clearer local arcs, slow down, stop or enter constrained creep recovery.

System Overview

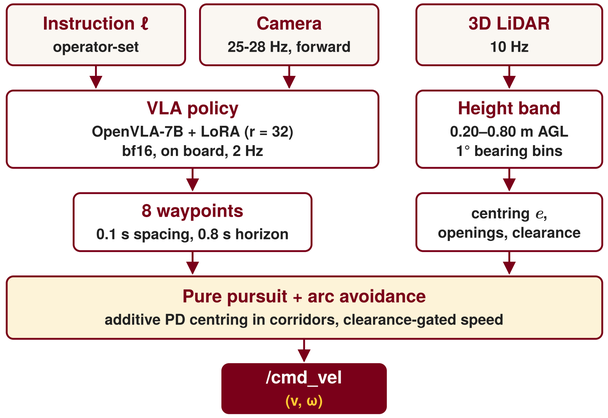


Fig. 1. Two tracks meet at the controller, and nothing leaves the robot.

1 On-Board Policy

One RGB frame and a standing instruction produce eight waypoints in the robot body frame. Waypoint-based navigation is related to ViNT [3]:

$$\pi_\theta : (I_t,\ \ell) \rightarrow (p_1,\ \ldots,\ p_8) \in \mathbb{R}^{16}$$

The outputs are cumulative positions over a 0.8 s horizon, not per-step deltas. OpenVLA is adapted with an OFT action head and rank-32 LoRA [1, 2]. The controller uses path shape, rather than the predicted speed.

The merged policy runs on board in bf16 at about 2 Hz. The OFT head predicts all eight waypoints in parallel; a separate 20 Hz loop applies control and LiDAR overrides.

2 Single-Scan LiDAR Geometry

One scan, no map, no accumulation. Points are kept in a band at robot height, then reduced to one minimum range per 1° bearing bin.



Fig. 2. Recycling bin beside the robot.

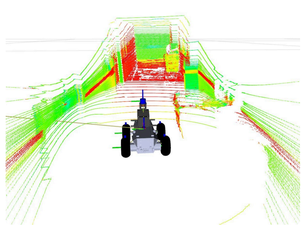


Fig. 3. Robot and its LiDAR point cloud.

Centring. With d_L, d_R the median side-sector ranges,

$$e = (\ d_\mathrm{R} - d_\mathrm{L}\ ) \ / \ 2$$

is the offset from the centreline, positive to the left — meaningful only between two walls, so it is held off at junctions.

Openings. A run of bins beyond the open range counts as a passage only if it is wide enough for the robot at its depth. Runs split at an interior dip deep relative to its flanking peaks, so a wall between two branches separates them.

Labels. Each opening is named by bearing — forward within ±25°, left or right beyond it — and the scene is called a corridor, a branch, a T-junction or a dead end only once the same label holds for five consecutive scans.

Band. A floor offset of -0.66 m in the robot frame was measured from LiDAR scans and used as a fixed calibration. The filter keeps points 0.20-0.80 m above the floor, not above the sensor. Obstacles outside this band, such as floor cables and overhangs, are not represented.

No accumulated map. Obstacle geometry is rebuilt each scan; scene-label filtering, derivative smoothing and recovery state persist. Occluded branches remain unseen until the robot is nearby, so LiDAR confirms rather than anticipates a junction.

3 Control and Safety

Pure pursuit. The first waypoint past the lookahead distance fixes the unique arc through the origin tangent to the heading [4]:

$$\kappa = \frac{2 y_\mathrm{L}}{{x_\mathrm{L}}^2 + {y_\mathrm{L}}^2}, \qquad \omega = v\ \kappa$$

Additive centring. In corridors, a LiDAR-based PD correction is added to policy steering; it is disabled at junctions. **Arc avoidance.** Candidate arcs are scored by free length and deviation from the policy, with a penalty for switching sides.

Clearance gate. Arc speed scales over 3.0-1.0 m; below 1.0 m, stop or creep if a clearer arc exists. Emergency lane <0.40 m: stop.

Clear	Slow	Stop
≥ 3.00 m full speed	3.00 to 1.00 m stop or creep below	< 0.40 m emergency lane

4 Obstacle Avoidance

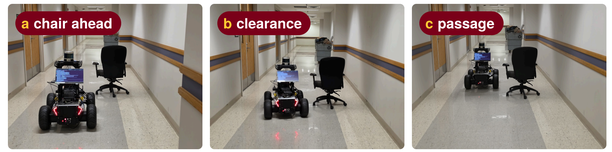


Fig. 4. (a) The robot passes left of a chair. (b) LiDAR points guide a swept-band arc search. (c) Passage alongside the chair. The method is related to reactive arc-based avoidance [5], not a full dynamic-window implementation.

Decision vs. Timing

Open-loop timing test. With a turn instruction already active, predicted endpoints remain near straight before the demonstrated turn.

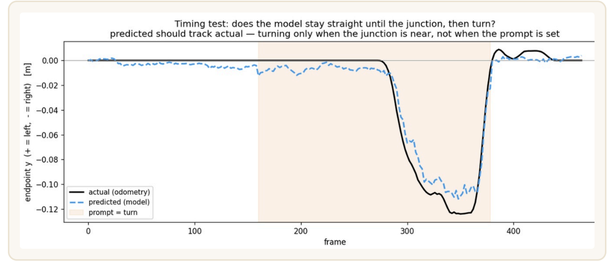


Fig. 5. Open-loop endpoint offset vs. odometry. Shaded: turn instruction active.

1. **Hold.** The endpoint stays near zero well after the instruction is set, while the corridor ahead is unchanged.
2. **Turn.** Predictions deflect near the driver's turn in this recorded sequence; this is not a closed-loop success measure.
3. **Return.** Predictions return toward zero as the demonstrated turn ends.

Platform



Base	AgileX Hunter 2.0, Ackermann, ≈1.6 m turning radius
Sensing	3D LiDAR at 10 Hz; one forward RGB camera
Compute	Jetson Orin on board, ROS 2 Humble
Rates	LiDAR 10 Hz; VLA ≈2 Hz; control 20 Hz

Conclusion

A 7B-class policy runs on board, producing waypoints from an image and an operator-set instruction. Single-scan LiDAR adds corridor centring, arc avoidance and speed gating without a map. The open-loop example suggests image-dependent turn timing; repeated closed-loop trials remain future work.

Future Work

- **Closed-loop evaluation:** success rate, minimum clearance and interventions over repeated runs, compared with the VLA alone.
- **People in motion:** track people across scans so the robot can anticipate someone walking toward it.
- **Recovery data:** off-centre and human-corrected driving, so the policy learns to recover, not only to follow clean demonstrations.
- **Fuller turns:** close the remaining gap between the predicted turn and the driven one.
- **More buildings:** new layouts and junction types beyond the one building tested so far.

Acknowledgements

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